



ĐẠI HỌC
BÁCH KHOA HÀ NỘI
HANOI UNIVERSITY
OF SCIENCE AND TECHNOLOGY

43rd HUST Annual Student Research Conference

A Collaborative Multi-agent Reinforcement Learning-Based Framework for Automated Related Work Summarization

ID: S.1.17

Students: Dang Hai Dang, Pham Bao Yen, Nguyen Quang Huy, Do Truong Giang

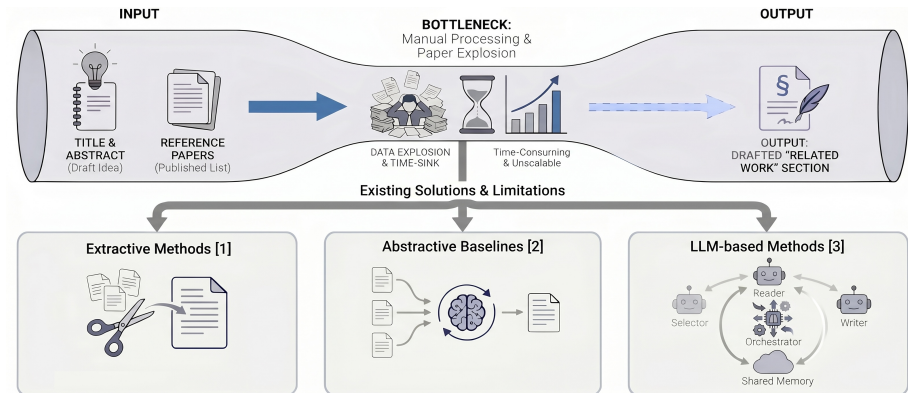
Supervisor: Prof. Huynh Thi Thanh Binh, Dr. Nguyen Thi Huong

Hanoi University of Science and Technology

May 13, 2026

ONE LOVE. ONE FUTURE.

Problem Statement



[1] Cong Duy Vu Hoang and Min-Yen Kan. "Towards automated related work summarization". In: *Coling 2010: Posters*. 2010, pp. 427–435

[2] Wen Xiao et al. "PRIMERA: Pyramid-based Masked Sentence Pre-training for Multi-document Summarization". In: *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*. 2022, pp. 5245–5263

[3] Xiaochuan Liu et al. "Select, Read, and Write: A Multi-Agent Framework of Full-Text-based Related Work Generation". In: *Findings of the Association for Computational Linguistics (2025)*. Accepted for publication

- ⇒ Formulate the RWS problem as a **Dec-POMDP** framework to optimize a routing policy without fine-tuning.
- ⇒ Introduce **Critique** for inter-agent evaluation, guided by a **balanced reward** mechanism.
- ⇒ Optimize a multi-agent policy with **IPPO** and **parameter sharing** for scalable coordination.
- ⇒ Conduct extensive experiments, achieving about **8% improvement** on LLM-based metrics.

Architecture

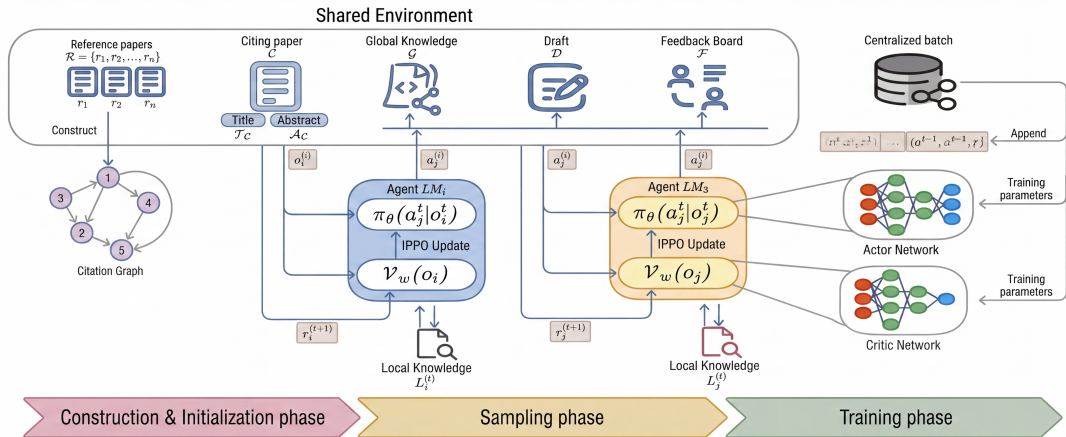


Figure 1: Proposed system architecture overview.

Experimental Setup and Results

► Experimental Setup

- Environment: Python 3.10, NVIDIA A100 GPU (Train & Infer: Qwen-32B-Instruct).
- Dataset: OARelatedWork [4].
- Metrics: LLM-as-a-Judge (inc. Coverage, Logic, Relevance) [3].
- Baselines: PRIMERA [2], GPT-4o, Select-RW [3].

► Results

Table 1. Effectiveness

Model	Cov.	Log.	Rel.	Avg.	Tok.
PRIMERA	2.48	2.71	3.02	2.64	186K
GPT-4o	3.18	4.08	3.96	3.36	2.1K
Select-RW	3.36	4.21	4.18	3.53	186K
CREW	3.62	4.39	4.31	3.77	153K

Table 2. Ablation Study

Config	Cov.	Log.	Rel.	Avg.	Tok.
w/o Draft	3.41 (-.21)	4.02 (-.37)	4.10 (-.21)	3.50 (-.27)	150K
w/o Crit.	3.35 (-.27)	4.08 (-.31)	4.02 (-.29)	3.42 (-.35)	143K
w/o BB.	3.29 (-.33)	3.95 (-.44)	3.98 (-.33)	3.35 (-.42)	160K
CREW	3.62	4.39	4.31	3.77	153K

[4] Martin Docekal, Martin Fajcik, and Pavel Smrz. "Oarelatedwork: A large-scale dataset of related work sections with full-texts from open access sources". In: *arXiv preprint arXiv:2405.01930* (2024)

[2] Wen Xiao et al. "PRIMERA: Pyramid-based Masked Sentence Pre-training for Multi-document Summarization". In: *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*. 2022, pp. 5245–5263

[3] Xiaochuan Liu et al. "Select, Read, and Write: A Multi-Agent Framework of Full-Text-based Related Work Generation". In: *Findings of the Association for Computational Linguistics* (2025). Accepted for publication

► Conclusions

- **Collaborative Framework:** Proposed CREW, a MARL-based system for automated related work summarization.
- **Empirical Success:** Improved coverage, relevance, and logic through dynamic agent coordination.

► Future Work

- **System Expansion:** Integrate external database crawling and dynamic online document graphs.

THANK YOU FOR LISTENING!

- [1] Cong Duy Vu Hoang and Min-Yen Kan. “Towards automated related work summarization”. In: *Coling 2010: Posters*. 2010, pp. 427–435.
- [2] Wen Xiao et al. “PRIMERA: Pyramid-based Masked Sentence Pre-training for Multi-document Summarization”. In: *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*. 2022, pp. 5245–5263.
- [3] Xiaochuan Liu et al. “Select, Read, and Write: A Multi-Agent Framework of Full-Text-based Related Work Generation”. In: *Findings of the Association for Computational Linguistics* (2025). Accepted for publication.
- [4] Martin Docekal, Martin Fajcik, and Pavel Smrz. “Oarelatedwork: A large-scale dataset of related work sections with full-texts from open access sources”. In: *arXiv preprint arXiv:2405.01930* (2024).